

SDI Programme Status

Three workstreams
· one calendar

ISSUED 27 AUG 2026

OWNERS J. GRAY · Y. KUMAR · M. YAZIR

COMPILED BY J. GRAY

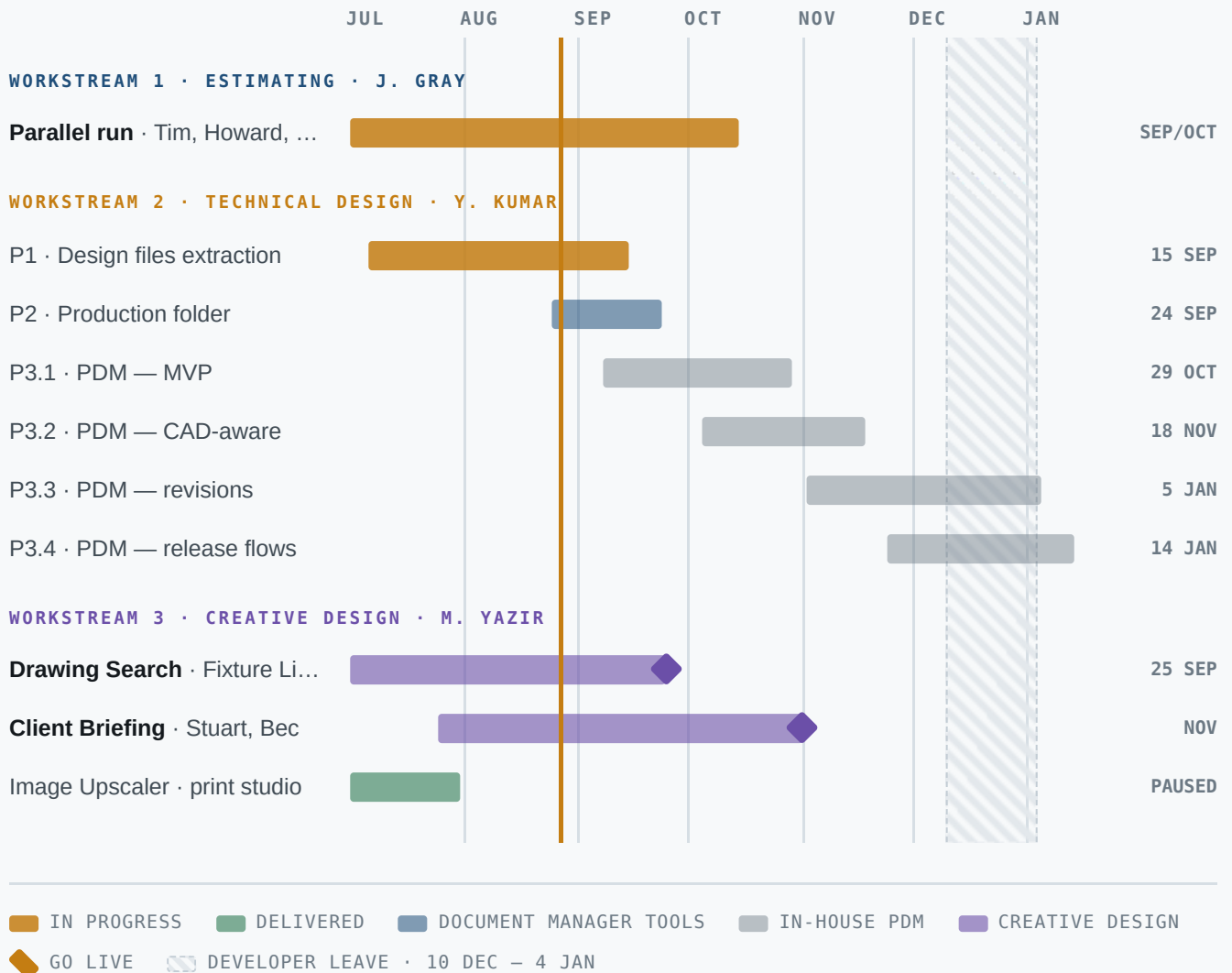
Status across **SDI Estimating Intelligence**, **SDI Technical Design Intelligence** and **SDI Creative Design Intelligence** in one document. They run **in parallel, not in sequence**, under three different owners, with no shared critical path — which is why they had never been drawn against the same calendar until now, and why doing so is the first thing in this report.

WORKSTREAM	WHAT IT DELIVERS	OWNER	NEXT DATE	STATE
1 · ESTIMATING	SDI Estimating Intelligence Drawings in, costed workbook out	J. Gray	Sep / Oct 26	PARALLEL RUN
2 · TECH DESIGN	SDI Technical Design Intelligence Document Manager tools, then an in-house PDM	Y. Kumar	15 Sep 26	P1 · 60%
3 · CREATIVE	SDI Creative Design Intelligence Drawing search, client briefing, print studio	M. Yazir	25 Sep 26	50% · 25%

01 The programme on one axis

Three workstreams, three owners, one calendar.

ALL WORKSTREAMS · 1 JUL 2026 → 31 JAN 2027



What one axis shows that three trackers did not

Four go-lives inside five weeks. Technical Design P1 on 15 Sep and P2 on 24 Sep, Drawing Search on 25 Sep, and the estimating engine targeting September / October. Four releases into the same business, from three owners who do not share a plan, inside the same month — and two of them land on consecutive days.

The first go-live is not the one everybody is watching. Drawing Search rolls out on 25 Sep, ahead of the estimating engine that every status report leads with.

Nothing blocks anything — worth stating plainly, because it is the good news. No workstream waits on another, so a slip in one does not move the others. The exposure is not dependency, it is **attention**: the same estimators and designers are the users for more than one of these, and all four land at once.

02

Workstream 1 — SDI Estimating Intelligence

STATUS	OWNER	USERS	
Parallel run	J. Gray	Tim · Howard · Tony	
			TARGET GO-LIVE SEPTEMBER / OCTOBER 2026 contingent on testing, adoption and holiday cover

Deterministic cost-estimating engine — SolidWorks PDF and DXF drawings in, a costed five-tab workbook out, mirroring the manual estimating workbook to the cell. All 80 workbook formulas reverse-engineered; a growing store of historical SDI jobs feeds comparables on material, labour and operations.

Where it is

In parallel run. The engine produces the workbook; the parity report compares it to the manually completed one cell by cell. What has changed this period is mostly about **trust** — several things reported success when they had not succeeded, and those now fail honestly.

A run that produces no workbook now says so FIXED

A job ran for sixteen minutes, reported *done*, and produced nothing. One missing template file silently took out the estimate, the quote, the job report, the Decision Report and the AI Provenance sheet at once, while the runner still reported success. The run now fails with an explanation, and the template is checked *before* the run starts.

The parity report shows what it actually compared FIXED

The HTML report was not always written; where it was, the variance column was calculated from a different pair of numbers than the difference column beside it. It is now written on every run, and every figure comes from the same source as the figure next to it.

Three files per comparison, not two — and the HTML one is the one to read.

The price chain, stated as it is wired DOCUMENTED

Seven sources, strictly ordered, first hit wins — no scoring across sources. **UDEF parts master** (exact code 0.95, fuzzy 0.82) → **ERP parts master** (purchased items, 0.55–0.88) → **bought-in catalogue** → **historical comparables** → **supplier catalogue** → **standard commodity** → **web / LLM** (capped 0.68).

The first six are reproducible; the seventh is not, which is why it is last and capped. **A zero is never a price** at any step.

A category word was being used as a part code FIXED

Raised by Tim. He can find `FIXING41`, `FIXING513` and `FIXING1122` in the parts master; he cannot find `FIXING`, because that is the word for what the thing *is*. The engine was asking the catalogue for it as though it were a code — which in the ERP parts master would have returned one price for every generic fixing on every drawing, at the highest confidence the chain can express. Class words are now refused as identifiers, and the line prices from its description.

Two things closed today DONE

The SDILive password is rotated and verified against the live catalogue, and the credential is out of the source tree — it had been a literal in seventeen tracked files, with the environment file itself tracked by git, so "put the new password in .env only" would have committed it straight back into the same exposure.

A permanent SolidWorks seat is installed on the estimating laptop. That unblocks four finished tools in Workstream 2, including the DXF flat-pattern generation this engine most needs — see below.

03 **Workstream 2 — SDI**
Technical Design Intelligence

STATUS	OWNER	RESOURCE
P1 · 60%	Y. Kumar	Sole developer

NEXT GO-LIVE
15 SEPTEMBER 2026
phases run in sequence, not in parallel

What it is, in one paragraph

Two threads. First the **Document Manager tools** — they read SolidWorks project data without needing a SolidWorks seat, and pull the right drawing pack for a job straight into the estimating app and into the shop-floor Production folder. Then an **SDI in-house PDM**, in four phases, giving engineering proper file management, revision control and release workflows.

This is **not** integration with the SOLIDWORKS PDM vault. SDI is building its own, and the Document Manager tools read the live CAD share *read-only*. Getting that wrong changes what the work is.

REF	PHASE	WHAT IT GIVES YOU	GO-LIVE	STATUS
P1	Design files extraction Document Manager API	Pulls the correct drawing pack and manufacturing files for a job straight into the estimating app. Serving the estimating app since 24 August.	15 Sep 26	60% DONE
P2	Production folder auto-extraction Document Manager API	Collects the shop-floor drawing pack on its own. Picks the latest revision by reading <i>inside</i> the file rather than trusting the filename, and reports near-misses instead of guessing.	24 Sep 26	NOT STARTED
P3.1	PDM — trustworthy files In-house, phase 1	Check-in and check-out, versioning, locking and history — so engineers can work on the same job without treading on each other.	29 Oct 26	NOT STARTED
P3.2	PDM — CAD-aware In-house, phase 2	The system understands how models link together: where a part is used, what references are missing, whether versions are consistent.	18 Nov 26	NOT STARTED
P3.3	PDM — engineering revisions In-house, phase 3	Revision control with approval: work in progress → pending approval → released, not approved, or obsolete.	5 Jan 27	NOT STARTED

REF	PHASE	WHAT IT GIVES YOU	GO-LIVE	STATUS
P3.4	PDM — release workflows In-house, phase 4	Separate release routes for work in progress, prototype and production components.	14 Jan 27	NOT STARTED

Built on GitLab and LFS with a Document Manager intelligence layer over PostgreSQL. Around 50 requirements captured from design, engineering and directors; the register is still open.

Why it runs into January, and it is not the amount of work

There is **one developer**. Requirement collection, development, unit testing, peer review, deployment and go-live are all hands-on and have to run one at a time — only user testing overlaps, because users do that. Add leave from 10 December to 4 January and the last two PDM phases land in the new year.

The dates are a capacity constraint, not an estimate of effort. If they need to come forward, the lever is a second pair of hands, not a tighter plan.

Four finished tools were held by a licence — the seat is now installed CHANGED

These were never planned work. They are **built and demoed to Ed Cooper and Ian B**, and every one carried *go-live subject to SolidWorks licence*. A permanent seat is now on the estimating laptop, so the constraint is lifted there; the remaining question is which machines get a seat, not whether the tools can run.

- **C1 · Multi-format export pack** — one click from a finished design to **DXF flat patterns**, PDF and DWG drawings, STEP per part and IGES per body.
- **C2 · Design validation suite** — topple, weight, interference and cross-part hole alignment in one pass. Topple checked on a live assembly at ~25.9°.
- **C3 · Drawing generation** — views, sheets, BOM and title block from a finished model; designers add the critical dimensions.
- **C4 · Machine tolerance checks** — against the real TRUMPF tool list and SDI standards. Still in proof of concept.

Why C1 matters most. The estimating engine's single largest source of inaccuracy is drawings arriving without flat patterns — with no DXF it has to infer geometry, and that is where the wrong numbers come from. C1 generates them, and it works today.

Behind the plan · research pipeline, no dates yet

PDF → 3D reconstruction POC PENDING

Rebuild SolidWorks parts and assemblies from PDF drawings — it would close the gap on every legacy job that has drawings but no model.

BOM to Excel QUICK WIN

Export a bill of materials from any model or drawing using Ian's custom property spec, normalised with anomalies flagged.

3DS Max automation POC PENDING

Drive 3DS Max through its API so the creative side feeds the same pipeline the engineering side uses.

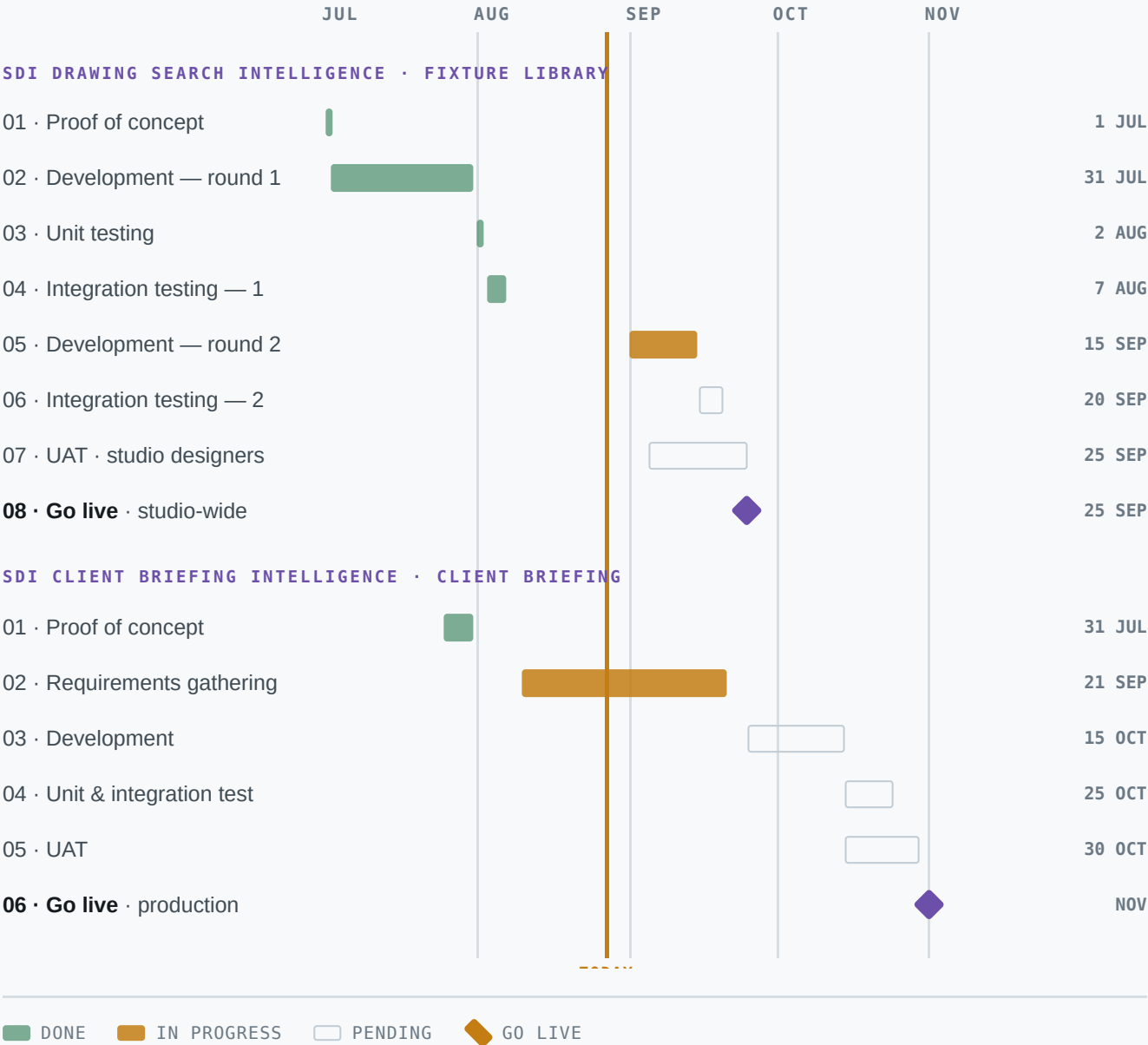
Electrical & AV schematics CONCEPT

Schematic layouts connecting power, cabling, LED strips, displays and drivers at the top level of a job.

04 Workstream 3 — SDI Creative Design Intelligence

STATUS		OWNER	PROJECTS		FIRST GO-LIVE	
In build		M. Yazir	2 live · 1 paused		~25 SEPTEMBER 2026	
					the earliest in the whole programme	
SERVICE NAME	TRACKER NAME	FOR	START	LIVE	DONE	
SDI Drawing Search Intelligence	Fixture Library	Ed, Stuart	1 Jul 26	~25 Sep 26	50%	
SDI Client Briefing Intelligence	Client Briefing	Stuart, Bec	31 Jul 26	~Nov 26	25%	
Image Upscaler	Image Upscaler	Sarah, Dean	1 Jul 26	Paused Aug 26	56%	

WORKSTREAM 3 · PHASE DETAIL · 1 JUL → 30 NOV 2026



Three overlaps worth knowing about before they bite

- **Drawing Search — UAT starts before the build it is testing is finished.** UAT runs 5–25 Sep; development round 2 runs to 15 Sep and integration round 2 to 20 Sep. Studio designers are on a build still being hardened, so early feedback may be against code that changes underneath them.
- **Drawing Search — go-live is the day UAT ends.** ~25 Sep, nothing between sign-off and studio-wide rollout. Anything found in the final week lands on the rollout date.
- **Client Briefing — four days between requirements sign-off and development.** Requirements end 21 Sep, development starts 25 Sep, and the scope document has to be agreed in between. The entire build is specified by it.

Client Briefing also runs unit & integration testing (15–25 Oct) and UAT (15–30 Oct) concurrently rather than in sequence — a deliberate compression on a client-facing release, worth knowing as the plan rather than discovering as a slip.

SDI Drawing Search Intelligence · phase detail

#	PHASE	START	END	DURATION	STATUS	NOTES
01	Proof of concept Stakeholder validation	1 Jul 26	1 Jul 26	1 day	DONE	Prototype demoed to & Stuart. Green light received.
02	Development — round 1 Full build	2 Jul 26	31 Jul 26	4 weeks	DONE	CLIP search PDF indexing GPU support Flask UI, sc cache.
03	Unit testing Internal — developer	1 Aug 26	2 Aug 26	2 days	DONE	Search accuracy, edge cases GPU fallback network safety.
04	Integration testing —	3 Aug 26	7 Aug 26	5 days	DONE	Finding: Renders

#	PHASE	START	END	DURATION	STATUS	NOTES
	round 1 Live data test					folders not indexed. Image count incomplete (~162k of ~268k). Ser back to dev
05	Development — round 2 Fixes & hardening	1 Sep 26	15 Sep 26	2 weeks	IN PROGRESS	Indexing ~446k remaining images (running). Wider rules security review, user guide.
06	Integration testing — round 2 Re-test after fixes	15 Sep 26	20 Sep 26	5 days	PENDING	Verify all ~268k images searchable. Confirm all models accurate.
07	UAT User acceptance testing	5 Sep 26	25 Sep 26	3 weeks	PENDING	Studio designers to test on real projects. Feedback + sign-off.
08	Go live Studio-wide rollout	~25 Sep 26	—	—	PENDING	IIS/nginx deploy. AD authentication index copy to K: drive.

What it actually does, since the name does not say

Image and drawing search for the design studio. **CLIP-based visual search** with PDF indexing over roughly 268,000 images, GPU-accelerated with a CPU fallback, behind a Flask UI with a scan cache. SDI has drawn the same bracket, the same shelf and the same plinth many times over, and until now the only index of that was what a designer happened to remember.

It is not the estimating engine's description matching pointed at a different table. That scores word overlap between two pieces of text; this searches the *images*, which is why it can find a fixture whose description never said what it looked like. The two may meet later — a drawing found here carries a part code, and a part code is a key the price chain already understands — but they are separate pieces of work today.

SDI Client Briefing Intelligence · phase detail

#	PHASE	START	END	DURATION	STATUS	NOTES
01	Proof of concept Stakeholder approval	25 Jul 26	31 Jul 26	1 week	DONE	POC presented and approved by Stuart & Ben. Proceed to full project.
02	Requirements gathering Stuart's large job, needs detail	10 Aug 26	21 Sep 26	6 weeks	IN PROGRESS	Detailed requirements sessions with Stuart. See doc — sign off before dev.
03	Development Full build	25 Sep 26	15 Oct 26	3 weeks	PENDING	Build all features from approved requirements.
04	Unit & integration	15 Oct 26	25 Oct 26	10 days	PENDING	Unit + integration tests. Bug

#	PHASE	START	END	DURATION	STATUS	NOTES
	testing Developer-led					fixes. Test report.
05	UAT User acceptance testing	15 Oct 26	30 Oct 26	2 weeks	PENDING	Client/student team tests real scenarios. Sign-off from Stuart & B
06	Go live Production release	~Nov 26	—	—	PENDING	Deploy to production. Handover, training if needed.

Image Upscaler

PAUSED AUG 26 · 56%

Print Studio — AI image upscaling and RGB → CMYK conversion (FOGRA39 ICC) for large-format print. Proof of concept, full build, unit testing and stakeholder review all complete and signed off; the colour bug found during build was fixed and **independently verified by Sarah** before handover. Review recorded CMYK ✓, TIFF ✓, workflow praised.

Paused because other studio priorities took precedence, not because anything failed — and the CMYK conversion it produced remains in active daily use. Restartable when capacity allows.

One date not taken as given

The Client Briefing go-live has been described in passing as **November 2027**. The tracker says **~Nov 2026**, and every date feeding it agrees — development ends 15 Oct 26 and UAT ends 30 Oct 26, so a 2027 release would leave a twelve-month gap after sign-off. **2026 is used throughout this report.** If 2027 is a deliberate hold it wants saying, because nothing else on the plan implies it.

Ordered by what it costs to leave alone. Recorded here rather than left to be rediscovered.

Read the balance of this list carefully

Most of what follows sits on Workstream 1, and that is **not** a statement about which workstream is in the best shape. It reflects how each is reported: Workstream 1 is reported from its own source code, where a defect can be found by looking; Workstreams 2 and 3 are reported from delivery trackers, which record phases and dates rather than faults. The two are not comparable, and a workstream with fewer entries here has not been shown to have fewer problems — only fewer that this method can see.

A published API secret has never been rotated ACT

The BrightHR client secret has been in the estimating repository's history since its first commit in April, and the repository was public for four months. It is now private and the environment files are untracked — **neither of those undoes history**. Anyone who has ever cloned it still has the value. Rotating the credential is the only cure. The SDILive database password has been rotated and verified; this one has not.

Workstream 1.

The new database password is not yet on the server OPEN

The laptop runs on the rotated login, verified against the live catalogue. SDI-APP01's two environment files still carry the old value and the service needs restarting after they are updated. Worth checking the SQL error log for failed logins a day or two afterwards, to find any consumer nobody remembered.

Workstream 1. One trap: the password is pasted into an ODBC connection string, where a semicolon ends the value and starts a new keyword — the server then answers *"Login failed"*, which reads as the wrong password rather than an unusable one. The engine now refuses such a password up front and names the character.

The server cannot report which build it is running [OPEN](#)

The health check shows `commit: unknown` on SDI-APP01, and parity runs there need one setting pointed at the right Python. A configuration script fixing both is written and **not yet run on the server**. This is the one host where two portals can drift apart, which is what the field exists to catch.

Workstream 1.

Three costing faults on the Dyson pack [OPEN](#)

- **Cross-job contamination** — a fixing from an unrelated fragrance-cabinet job was costed onto a Dyson estimate. The test guarding this was found to be vacuous and removed; the guard itself needs rebuilding.
 - **Prose read as a part number** — a draughtsman's name and date in a title block was read as a part, given a material from a revision note saying the panel had been *removed*, and costed anyway.
 - **Powder coating at £0.00** on a powder-coated job. The rate is settled — £9.73/kg, evidenced from the manual sheets on jobs 1303A and 1304; a second £12.50 constant was traced and is dead code with no callers. What remains open is policy: whether an unstated finish defaults to powder with a visible assumption flag, or is a required estimator decision.
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Workstream 1.

Two price sources are wired to tables that are empty [OPEN](#)

Rung 3 of the price chain reads the bought-in catalogue where rows are active. A migration moved that catalogue to a new table and set every migrated row inactive, so the rung returns nothing — the successor holds 44 rows. And the table where a priced estimate is meant to be saved **does not exist**: no run has ever recorded which source won a price.

Workstream 1. Two small decisions: repoint rung 3 at the successor table, and either create the results table or remove the save.

Size is not weighted in description matching [OPEN](#)

Historical price matching scores word overlap and has no idea which word is the size, so a wrong-size line of the same family can outscore the right part written with different punctuation. Recorded as a deliberately failing test so it cannot be forgotten. Fixing it changes matching for every priced line, so it wants doing on its own.

Workstream 1.

Supplier price lists are not loaded [OPEN](#)

The ingester is built and tested — it turns a supplier's spreadsheet into catalogue rows, reads the unit from the column then the price heading then the description and says which of the three it used, and refuses a file that is not a price list rather than half-parsing one. It needs the actual files: **Elite** (Tim Allen), **Eagle** (Cameron) and **Thermaset** (Donna).

Workstream 1.

Drawing Search go-live has no buffer [OPEN](#)

Studio-wide rollout on 25 Sep is the same day UAT ends, and UAT overlaps both remaining build phases. This is a schedule shape rather than a defect, and it is the single most likely place for the programme's first go-live to slip.

Workstream 3. A decision for the owner, not a fix.

Workstream 2 has one developer and no cover [OPEN](#)

Requirement collection, development, unit testing, peer review, deployment and go-live are all hands-on and run one at a time; only user testing overlaps. Leave from 10 December to 4 January sits inside the PDM phases. Every date from P3.1 onwards depends on one person being available, and there is no second pair of hands to move work to.

Workstream 2. A capacity constraint, not an estimate of effort — which means it cannot be recovered by re-planning.

The PDM requirement register is still open

OPEN

Around 50 requirements have been captured from design, engineering and directors, and the register has not been closed. The four PDM phases carry dates against a scope that can still grow.

Workstream 2.

The indexing finding has not yet been re-verified

OPEN

Integration testing round 1 found the Renders folders were not indexed — roughly 162,000 of 268,000 images. Round 2 (15–20 Sep) has to confirm all ~268,000 are searchable and all three models accurate. Until it does, the 25 Sep go-live rests on a fix that has been made but not yet proven, and it is the same test that caught the problem the first time.

Workstream 3.

Client Briefing is building to a document not yet signed

OPEN

Requirements gathering ends 21 September and development starts on the 25th. The scope document has to be agreed in those four days, and the entire build is specified by it. A week's delay in sign-off is a week off the front of a three-week build, not a week added to the end.

Workstream 3.

Which machines get a SolidWorks seat

OPEN

The estimating laptop now has one, which lifts the constraint on C1–C4 there. Nothing has been decided about the other machines — and it is the difference between four finished tools running for one person and running for the department.

Workstreams 1 and 2.

DWG drawings are silently ignored

PARKED

No converter is configured, so a DWG in a pack contributes nothing and says nothing. Parked deliberately, recorded so it is not rediscovered as a surprise.

Workstream 1.

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